YEAR 10 PHYSICS, AQA GCSE, PART 1 OF 3

PARTICLE MODEL OF MATTER

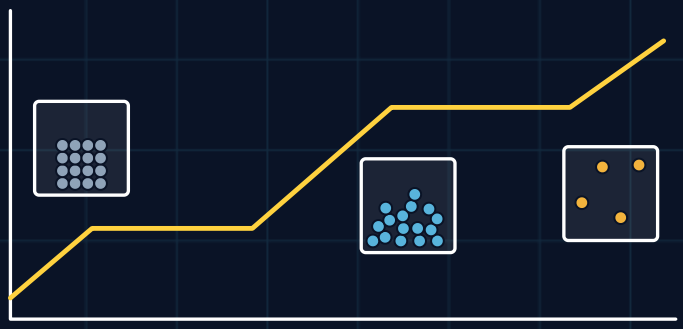
PART 1 OF 3

LESSONS 1–2

IN THIS BOOKLET

- Lesson 1: Solids, liquids and gases
- Lesson 2: Gas particles and pressure

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

HOW THIS BOOKLET WORKS

Everything around you is made of tiny particles. This unit uses that one idea to explain density, melting and boiling, why some things are hard to heat up, and why a gas pushes on its container. Keep the booklet: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations and diagrams. Fill in the slide examples with your teacher, then try “Your turn” alone.

TRY IT
Practice that builds up, then a challenge. Every second lesson ends with the slides' Quick check.

PRACTICAL
Record your measurements and work out results from them.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an “I can” check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: The particles in a solid are **close together** **(close together / far apart)**. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The unit of mass is **the newton** | **the kilogram** | **the joule** .

THE ROUTE THROUGH THE UNIT (THIS BOOKLET HIGHLIGHTED)

01 **Solids, liquids and gases** Slides 3–7

02 **Gas particles and pressure** Slides 8–12

03 Density

04 Measuring density (required practical)

05 Internal energy and heating curves

06 Specific latent heat

07 Specific heat capacity

08 Measuring specific heat capacity (required practical)

The **equation toolkit** near the back has every equation you have met so far, with formula triangles to help you rearrange. Teaching deck: **panphy.app/gcsephy/decks/particle-model**

Lesson

01 SOLIDS, LIQUIDS AND GASES

AQA Physics 4.3.1.1, 4.3.1.2; Combined Science 6.3.1.1, 6.3.1.2

BIG QUESTION **An ice cube, a puddle and the steam from a kettle are all water. What is different about them?**

BY THE END YOU CAN

- draw particle diagrams for a solid, a liquid and a gas
- describe how the particles are arranged and how they move
- name the six changes of state
- explain why mass is conserved when a substance changes state

particle **solid** **liquid** **gas** **change of state** **physical change** **conserved**

YOU WILL NEED

Demonstration: two sealed syringes (one of air, one of water); a beaker of ice on a balance, covered with cling film.

Teaching deck, slides 3–7
panphy.app/gcsephy/decks/particle-model/#3

REMEMBER

WHAT DO YOU ALREADY KNOW?

From Key Stage 3 and Year 9. Have a go.

1
Name the three states of matter.

Solid, liquid and gas.

2
Name the change of state when ice turns to water.

Melting.

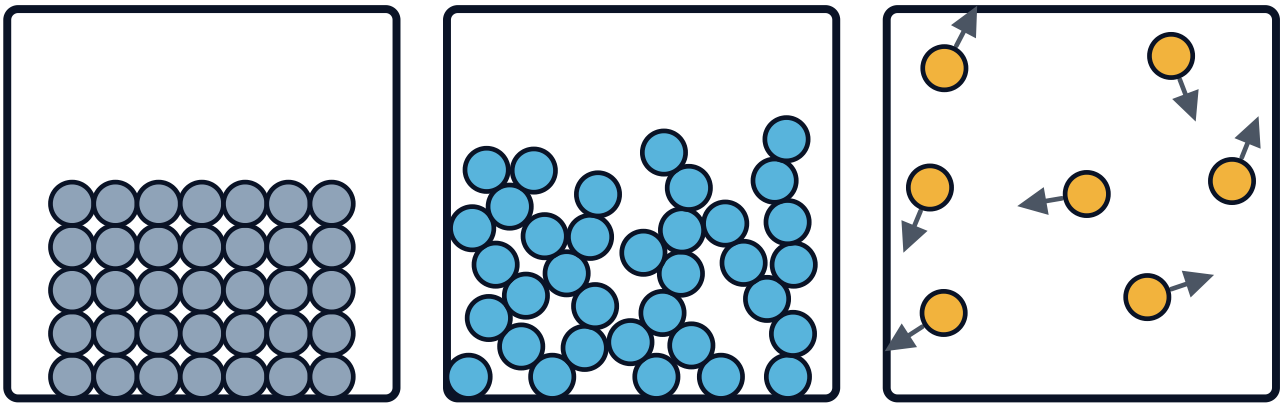
3
Name the instrument used to measure mass.

A balance.

EXPLAIN

ONE MODEL, THREE STATES

Everything is made of tiny **particles** (atoms or molecules), far too small to see. The **particle model** says that a solid, a liquid and a gas of the same substance contain the *same particles*. Only two things differ: how the particles are **arranged** and how they **move**.



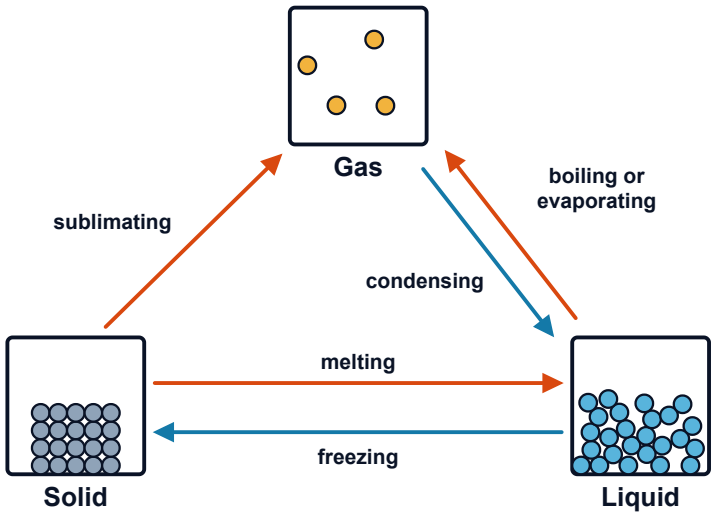
Solid Liquid Gas

Each circle is one particle. In a particle diagram, draw the circles the same size in all three states.

	SOLID	LIQUID	GAS
Arrangement	very close together, in a regular pattern	close together, in a random arrangement	far apart, in a random arrangement
Motion	vibrate about fixed positions	move around, sliding past each other	move quickly in all directions
Shape and volume	fixed shape, fixed volume	takes the shape of its container, fixed volume	fills its container
Can it be squashed?	no: the particles are already touching	no: the particles are already touching	yes: there is empty space between the particles

Get the marks: Draw a solid with the particles **touching**, in neat rows. Draw a liquid with the particles still **touching**, but in no pattern. Leaving gaps in a liquid is the most common mistake in the exam.

CHANGES OF STATE



Orange arrows: the substance is heated. Blue arrows: it is cooled.

A liquid can turn into a gas in two ways. **Boiling** happens only at the boiling point, with bubbles all through the liquid. **Evaporating** happens at any temperature, from the surface only. **Sublimating** is when a solid turns straight into a gas: solid carbon dioxide ("dry ice") does this.

MASS IS CONSERVED

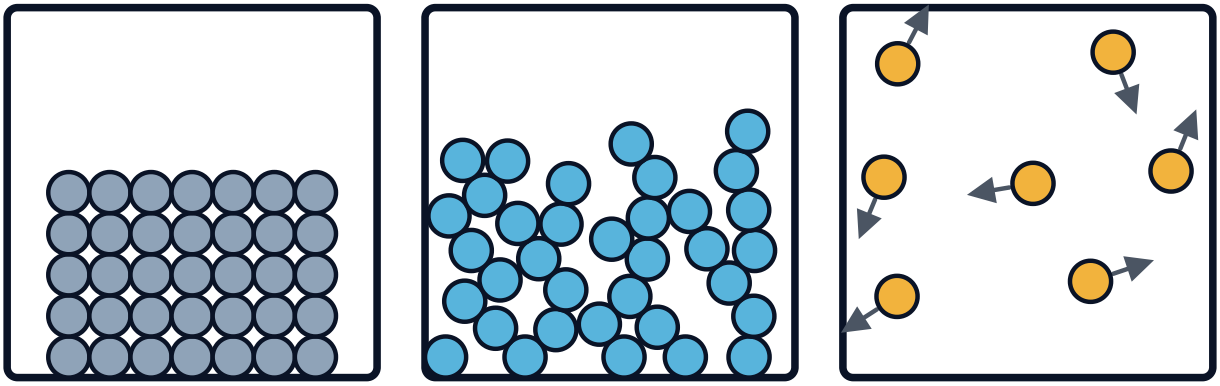
> ! When a substance changes state, **no particles are added or taken away**. They are only rearranged. So the mass stays the same: 50 g of ice melts to make 50 g of water.

A change of state is a **physical change**, not a chemical change. No new substance is made. If you reverse the change, the material gets its original properties back: melt ice and freeze it again, and you have ice again.

PRACTISE

TRY IT

1 The diagrams show the particles in a solid, a liquid and a gas. Write the state under each letter.



A is a solid (solid / liquid / gas). B is a liquid (solid / liquid / gas). C is a gas (solid / liquid / gas).

2 Draw a ring around the correct answer in each box.

The particles in a solid

move quickly in all directions	**vibrate about fixed positions**	slide past each other

The particles in a liquid are

far apart	in a regular pattern	**close together**

A gas can be squashed because its particles

are far apart	are soft	are small

3 SUPPORTED Complete the sentences.

When a solid is heated enough, it turns into a liquid. This is called **melting** **(melting / freezing / condensing)**.

When a gas is cooled enough, it turns into a liquid. This is called **condensing** **(boiling / condensing / sublimating)**.

When a solid turns straight into a gas, it is called **sublimating** **(evaporating / freezing / sublimating)**.

4 Give two differences between boiling and evaporating.

Boiling happens only at the boiling point; evaporating happens at any temperature.

Boiling happens throughout the liquid (bubbles); evaporating happens only at the surface.

5 A sealed bag holds 120 g of water. It is put in a freezer until all the water is ice. State the mass of the ice, and explain your answer.

120 g. The number of particles has not changed: none were added or removed, they were only rearranged.

6 CHALLENGE Liquid water can be poured, but a block of ice cannot. Use the particle model to explain why.

In a liquid the particles can move around and slide past each other, so the liquid flows.

In a solid the particles are held in fixed positions and can only vibrate.

INVESTIGATE

SEE IT: SQUASH IT, MELT IT

Your teacher will demonstrate. Write what you see.

Part 1. Your teacher blocks the end of a syringe of air and pushes the plunger. Then the same with a syringe of water.

7 Complete the table.

In the syringe	Can it be squashed?	What this tells you about the particles
air (a gas)	**yes, easily**	**there is a lot of empty space between them**
water (a liquid)	**no**	**they are already touching**

Part 2. A covered beaker of ice is put on a balance. The mass is read again when all the ice has melted.

8 Record the two readings and write a conclusion.

	Mass of beaker and contents / g
all ice	**186.4**
all water	**186.4**

The mass does not change when the ice melts, because the number of particles is the same.

9 Why must the beaker be covered?

So that no water evaporates and leaves the beaker, and no water from the air condenses on the cold ice.

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: “Compare” means say something about **both**. Give the arrangement and the motion for each state.

10 Compare the arrangement and the motion of the particles in a solid with those in a gas. **[4 marks]**

Solid: particles very close together (1) in a regular pattern, vibrating about fixed positions (1). Gas: particles far apart (1), moving quickly in random directions (1).

11 A student melts 25 g of solid wax in a sealed tube. **[3 marks]**

(a) State the mass of the liquid wax.

(b) Explain why melting is a physical change.

(a) 25 g (1). (b) No new substance is made (1); the wax recovers its original properties if the change is reversed, by cooling it (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Solid: close together, regular pattern, vibrating about fixed positions.
- Liquid: close together, random arrangement, moving past each other.
- Gas: far apart, random arrangement, moving quickly in all directions.
- Changes of state are physical changes. The mass is conserved because the number of particles stays the same.

COVER THE PAGE AND ANSWER

1 Describe the arrangement of the particles in a liquid.

2 Describe the motion of the particles in a solid.

3 Name the change of state from gas to liquid.

4 40 g of water boils away completely. What mass of steam is made?

I CAN…

- draw and recognise particle diagrams for a solid, a liquid and a gas
- describe the arrangement and motion of the particles in each state
- name the changes of state
- explain why mass is conserved in a change of state
- explain why a change of state is a physical change

Lesson

02 GAS PARTICLES AND PRESSURE

AQA Physics 4.3.3.1; Combined Science 6.3.3.1

BIG QUESTION **An aerosol can says “Do not expose to temperatures above 50 °C”. What could go wrong?**

BY THE END YOU CAN

- describe the motion of the molecules in a gas
- link the temperature of a gas to the average kinetic energy of its molecules
- explain how a gas pushes on the walls of its container
- explain why heating a gas at constant volume increases its pressure

random motion **kinetic energy** **temperature** **pressure** **collision** **constant volume**

YOU WILL NEED

Demonstration: a sealed round-bottomed flask joined to a pressure gauge, a large beaker or water bath, a kettle, a thermometer.

Teaching deck, slides 8–12
panphy.app/gcsephy/decks/particle-model/#8

REMEMBER

DO NOW

From earlier lessons. No notes!

1
Describe the arrangement of the particles in a gas.

Far apart, in a random arrangement.

2
Why can a gas be squashed easily?

There is a lot of empty space between its particles.

3
Which energy store does a moving object have? Write the equation for it.

The kinetic store.

$E_k = \frac{1}{2} m v^2$

EXPLAIN

A GAS NEVER KEEPS STILL

The molecules of a gas are in **constant random motion**. “Random” means that they move in all directions and at a range of speeds. They travel in straight lines until they hit each other or a wall, then bounce off.

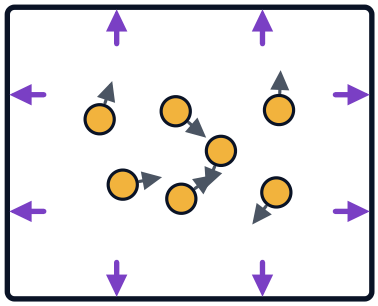
TEMPERATURE AND KINETIC ENERGY

> ! The **temperature** of a gas is related to the **average kinetic energy** of its molecules. Heat a gas and its molecules move faster: the hotter the gas, the greater their average kinetic energy.

We say *average* because at any moment some molecules are moving fast and some slowly.

WHAT CAUSES GAS PRESSURE?

1. The molecules **collide** with the walls of the container.
2. Each collision gives the wall a tiny **push** (a force).
3. There are billions of collisions every second, on every part of the wall.
4. Together they make a steady force on each square metre of wall: the **pressure** of the gas.

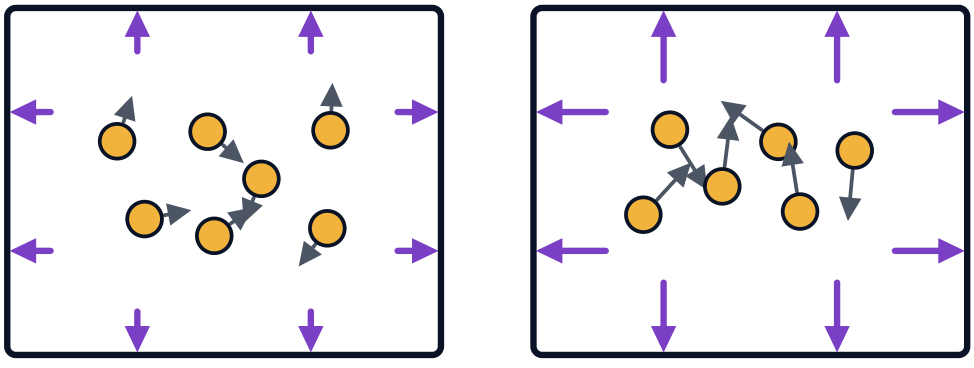


gas in a sealed box

→ push of the gas on the wall

HEATING A GAS AT CONSTANT VOLUME

"Constant volume" means that the container is sealed and cannot get bigger, like a steel can. Follow the chain of reasoning:



cooler gas **hotter gas, same box**

→ push of the gas on the wall

1. The gas is heated, so the average kinetic energy of the molecules increases: they move **faster**.
2. They hit the walls **more often**.
3. Each collision pushes on the wall with a **bigger force**.
4. So the **pressure increases**.

> **Get the marks:** The molecules do **not** get bigger when a gas is heated, and they do not "expand". They move faster. Never write that particles expand.

Cooling does the opposite: the molecules slow down, hit the walls less often and less hard, and the pressure falls.

PRACTISE

TRY IT

1 Draw a ring around the correct answer in each box.

The molecules of a gas move

in circles	**in random directions**	all in the same direction

When a gas is heated, its molecules

get bigger	**move faster**	move closer together

The temperature of a gas is related to the molecules'

average kinetic energy	size	mass

2 SUPPORTED Complete the sentences.

Gas pressure is caused by molecules **colliding with** (colliding with / sticking to / passing through) the walls of the container.

When a gas in a sealed can is cooled, the pressure **decreases** (increases / decreases / stays the same).

3 A gas in a sealed steel can is heated. Put the statements in order, 1 to 4, to explain what happens.

3	Each collision with the wall exerts a bigger force.
1	The average kinetic energy of the molecules increases, so they move faster.
4	The pressure of the gas increases.
2	The molecules hit the walls more often.

Order: 3, 1, 4, 2 (reading down the list).

4 A car is driven for a long time on a hot day. Explain why the pressure of the air in its tyres increases. Assume the volume of a tyre does not change.

The air in the tyre gets hotter, so the molecules move faster (their average kinetic energy increases). They collide with the tyre wall more often and with more force, so the pressure increases.

5 **CHALLENGE** A sealed bottle of air is put in a freezer. Mia says, “The pressure falls because some of the air particles disappear.” Explain why Mia is wrong, and give the correct reason.

The bottle is sealed, so the number of particles stays the same. The pressure falls because the molecules move more slowly: they hit the walls less often and with less force.

INVESTIGATE

SEE IT: PRESSURE AND TEMPERATURE

Your teacher will demonstrate. The flask is sealed, so its volume stays the same.

A sealed flask of air is joined to a pressure gauge and put in water at different temperatures. Record the readings.

6 Record the results.

Temperature of the air / °C	Pressure / kPa
20	101
40	108
60	115
80	122

7 Describe the pattern in the results.

As the temperature of the gas increases, its pressure increases.

8 Which two things were kept the same in this demonstration?

The volume of the gas (the flask is rigid) and the mass (or number of particles) of gas, because the flask is sealed.

CHECKPOINT

QUICK CHECK

Slide 12, Lessons 1–2. Answer each question, then check it as your teacher reveals the answer.

1

The particles are close together, in a random arrangement. Which state is this?

Liquid.

2

How do the particles in a solid move?

They vibrate about fixed positions.

3

A gas turns into a liquid. Name the change of state.

Condensing.

4

80 g of wax melts. What is the mass of the liquid wax?

80 g: the number of particles is the same.

5

What causes the pressure of a gas?

Its molecules colliding with the walls.

6

A sealed can of gas is cooled. What happens to the pressure?

It decreases: the molecules move more slowly.

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: A 3-mark “explain” about gas pressure needs three linked steps: **faster**, **collisions**, **pressure**.

9 An aerosol can is left in sunlight and gets hot. Explain, in terms of particles, why the pressure of the gas in the can increases. **[3 marks]**

The molecules gain kinetic energy and move faster (1). They collide with the walls of the can more often (1) and with a greater force, so the pressure increases (1).

10 Describe the motion of the molecules in a gas. **[2 marks]**

They are in constant motion (1), in random directions (and with a range of speeds) (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Gas molecules are in constant random motion.
- The temperature of a gas is related to the average kinetic energy of its molecules.
- Gas pressure is caused by the molecules colliding with the walls.
- Heating a gas at constant volume increases its pressure: faster molecules hit the walls more often and harder.

COVER THE PAGE AND ANSWER

1 What does “random motion” mean?

2 What causes the pressure of a gas?

3 A gas is cooled at constant volume. What happens to its pressure?

4 What is the temperature of a gas related to?

I CAN…

- describe the random motion of gas molecules
- link temperature to the average kinetic energy of the molecules
- explain how collisions with the walls cause pressure
- explain why heating a gas at constant volume increases its pressure

REFERENCE

KEY WORDS FROM THIS BOOKLET

Learn the meaning and the spelling. In exams, precise words earn marks: say “the particles vibrate about fixed positions”, not “the particles shake”.

Particle model L1: The idea that all matter is made of tiny particles, whose arrangement and motion explain how solids, liquids and gases behave

Change of state L1: Melting, freezing, boiling, evaporating, condensing or sublimating

Sublimating L1: Changing straight from a solid to a gas

Physical change L1: A change in which no new substance is made; the material recovers its original properties if the change is reversed

Conserved L1: Staying the same in total: mass is conserved in a change of state

Random motion L2: Movement in all directions, at a range of speeds

Gas pressure L2: The force on each square metre of a surface, caused by gas molecules colliding with it

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

LESSON 1: SOLIDS, LIQUIDS AND GASES

DO NOW

1. Solid, liquid and gas.
2. Melting.
3. A balance.

COVER THE PAGE

1. Close together, in a random arrangement.
2. They vibrate about fixed positions.
3. Condensing (condensation).
4. 40 g

LESSON 2: GAS PARTICLES AND PRESSURE

DO NOW

1. Far apart, in a random arrangement.
2. There is a lot of empty space between its particles.
3. The kinetic store. $E_k = \frac{1}{2}\,m\,v^2$

COVER THE PAGE

1. The molecules move in all directions, at a range of speeds.
2. Its molecules colliding with the walls of the container.
3. It decreases.
4. The average kinetic energy of its molecules.